

Part A: Order of Operations

Evaluate each expression completely.

1. $(-8 + (-19))^2 \div (-9)$

2. $(-9 + (-15))^2 \div (-8)$

3. $(-8 + (-15))^2 \div (-1)$

4. $-5 - 10 \cdot (7 + (-5))$

Part B: Fractions

Evaluate. Write every answer in lowest terms.

1. $3\frac{1}{2} + 1\frac{2}{3}$

2. $3\frac{1}{3} \div \frac{1}{12}$

3. $3\frac{2}{5} - 5\frac{1}{2}$

Part C: Proportions and Unit Rates

Solve each proportion for x , or find the unit rate.

1. $\frac{8}{x} = \frac{16}{28}$

2. $\frac{x}{14} = \frac{24}{56}$

3. Liam pays \$40 for 8 notebooks. Find the price per notebook.
4. Option A: \$22 for 11 juice boxes. Option B: \$150 for 10 juice boxes. Which option is the better price per juice box?

Part D: Classifying Numbers

Classify each number using all that apply - rational, irrational, whole, integer.

1. $3\frac{5}{6}$

2. -31

3. $\sqrt{32}$

Part E: Exponent Rules

Simplify. Write answers with positive exponents.

1. $9x^2 \cdot 5x^4$

2. $(9x^9)^3$

3. $6 \cdot 7^{-3}$

Part F: Roots

Evaluate or simplify each root.

1. $-\sqrt{1444}$

2. $\sqrt[3]{91125}$

3. $\sqrt{416}$

Part G: Percent Proportion and Estimation

Solve the percent proportion, or find each value.

1. $\frac{17/2}{x} = \frac{25}{100}$

2. Estimate 48% of 53 by rounding to 50% of 50.

3. 5% of 165

Part H: Percent Applications

Solve each percent application problem.

1. from 25 to 45

2. A store marks up a jacket that costs \$100 by 40%. Find the new price.

3. A recipe calls for 50 grams of flour. A student measures 35 grams. Find the percent error.

4. Ethan earns a 25% commission on sales of \$1580. Find Ethan's commission.

5. Ines's bill comes to \$40. Including a 5% tip, find the total amount Ines pays.

Part I: Linear Expressions

Simplify, combine, or factor each expression.

1. $11x + 7 - 5x + 3$
2. $11(12x - 6) + 5x + 5$
3. $(7x - 3) + (10x - 2)$
4. $-21x + 24$

Part J: Solving Equations

Solve each equation for x . Show your steps.

1. $9x + 8 = 17$
2. $6x + 9 = -39$
3. $9(7x - 7) + 4 = 571$
4. $8(5x + 7) + 11 = -293$

5. $-4x - 11 = -7x - 17$

6. $-11x + 6 = -10x - 1$

Part K: Solving Inequalities*Solve each inequality for x .*

1. $-6x + 8 < -22$

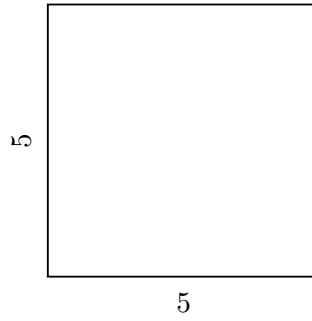
2. $6x + 4 < -20$

3. $7x + 6 > x + 30$

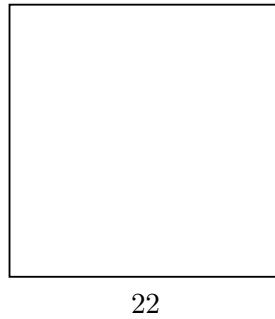
4. $-6x - 2 < 9x - 122$

Part L: Area*Find the area of each figure.*

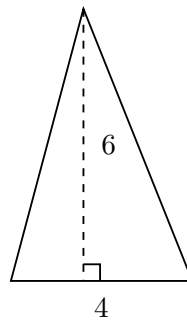
1. All measurements are in ft.



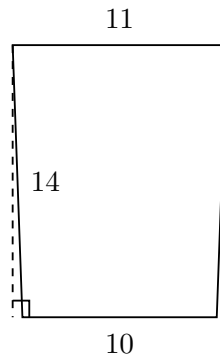
2. All measurements are in m.



3. All measurements are in cm.



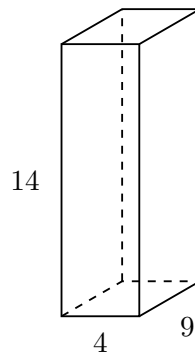
4. All measurements are in cm.



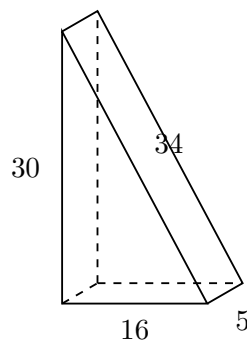
Part M: Volume of Prisms

Find the volume of each prism.

1. A rectangular prism. All measurements are in cm.



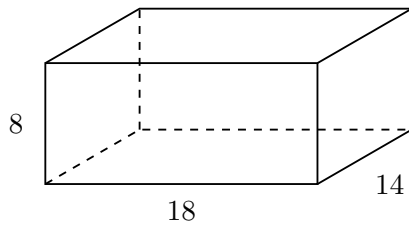
2. A triangular prism with a right-triangle base. All measurements are in in.



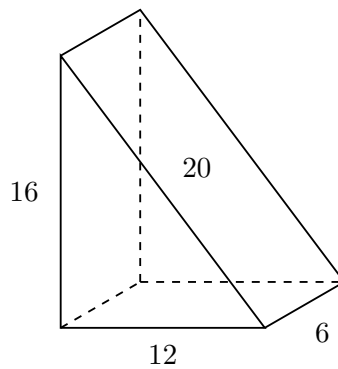
Part N: Surface Area of Prisms

Find the surface area of each prism.

1. A rectangular prism. All measurements are in m.



2. A triangular prism with a right-triangle base. All measurements are in cm.



Answer Key

Part A: Order of Operations

1. -81
2. -72
3. -529
4. -25

Part B: Fractions

1. $5\frac{1}{6}$
2. 40
3. $-2\frac{1}{10}$

Part C: Proportions and Unit Rates

1. $x = 14$
2. $x = 6$
3. \$5 per notebook
4. Option A

Part D: Classifying Numbers

1. Rational
2. Rational, Integer
3. Irrational

Part E: Exponent Rules

1. $45x^6$
2. $729x^{27}$
3. $\frac{6}{343}$

Part F: Roots

1. -38
2. 45
3. $4\sqrt{26}$

Part G: Percent Proportion and Estimation

1. $x = 34$
2. 25
3. $\frac{33}{4}$

Part H: Percent Applications

1. 80%
2. 140 dollars
3. 30
4. 395 dollars
5. 42 dollars

Part I: Linear Expressions

1. $6x + 10$
2. $137x - 61$
3. $17x - 5$
4. $3(-7x + 8)$

Part J: Solving Equations

1. $x = 1$
2. $x = -8$
3. $x = 10$
4. $x = -9$
5. $x = -2$
6. $x = 7$

Part K: Solving Inequalities

- | | |
|-------------|------------|
| 1. $x > 5$ | 3. $x > 4$ |
| 2. $x < -4$ | 4. $x > 8$ |

Part L: Area

- | | |
|-----------------|------------------|
| 1. 25 square ft | 3. 12 square cm |
| 2. 484 square m | 4. 147 square cm |

Part M: Volume of Prisms

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|-----------------|------------------|
| 1. 504 cubic cm | 2. 1200 cubic in |
|-----------------|------------------|

Part N: Surface Area of Prisms

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|------------------|------------------|
| 1. 1016 square m | 2. 480 square cm |
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